CS101 updated VU Midterm Past Paper MCQs Questions From 2020 to date
Created by APEX Team

What components handle the coordination of processes in an operating system's kernel?

a. File manager and memory manager

b. Scheduler and dispatcher

c. Process table and memory area

d. Interrupt handler and timer circuit

Correct Answer: b. Scheduler and dispatcher

What information is stored in the process table by the scheduler?

a. User's name and password

b. Process's priority and memory area

c. Mass storage operation status

d. Time slice duration

Correct Answer: b. Process's priority and memory area

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When is a process considered "waiting" according to the paragraph?

a. When it is in a state where progress can continue

b. When it is waiting for a time slice

c. When it is in a state of suspension

d. When it is interrupted by another process

Correct Answer: c. When it is waiting for an external event

What initiates the end of a time slice in a time-sharing/multitasking system?

a. Process completion

b. User intervention

c. Timer circuit generating an interrupt

d. Dispatcher's decision

Correct Answer: c. Timer circuit generating an interrupt

What is the procedure called when changing from one process to another in a time-sharing system?

a. Process termination

b. Process initiation

c. Time slice completion

d. Process switch or context switch

Correct Answer: d. Process switch or context switch

Which component of the CPU is responsible for reacting to interrupt signals?

a. Memory manager

b. Scheduler

c. Interrupt handler

d. Dispatcher

Correct Answer: c. Interrupt handler

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What is the primary purpose of a semaphore in an operating system?

a. Controlling access to printer devices

b. Managing process priorities

c. Enabling multitasking

d. Preventing memory overflows

Correct Answer: a. Controlling access to printer devices

How is deadlock defined in the paragraph?

a. When a process is interrupted

b. When two processes compete for resources

c. When a process cannot complete its task

d. When resources are requested partially and cannot be forcibly retrieved

Correct Answer: b. When two processes compete for resources

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What are the conditions that must be satisfied for deadlock to occur?

a. 1 and 2

b. 2 and 3

c. 1, 2, and 3

d. None of the above

Correct Answer: c. 1, 2, and 3

What is the technique called when non-sharable resources are made to appear sharable in an operating system?

a. Spooling

b. Multiprogramming

c. Deadlock detection

d. Privilege mode

Correct Answer: a. Spooling

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What is the role of auditing software in computer security?

a. Allocating resources to processes

b. Detecting unauthorized access

c. Managing memory limits

d. Controlling printer access

Correct Answer: b. Detecting unauthorized access

What is the term used to describe the situation in which an intruder tricks the operating system into allowing access beyond a user's privileges?

a. Privilege escalation

b. Memory limit override

c. Resource allocation

d. Deadlock

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Correct Answer: a. Privilege escalation

How is a CPU typically switched between privileged and non-privileged modes?

a. Using interrupt signals

b. With a "change privilege mode" instruction

c. By the operating system administrator

d. Automatically at system startup

Correct Answer: b. With a "change privilege mode" instruction

What can a single flaw in privilege level control lead to?

a. Unauthorized access to printer devices

b. Increased memory limits

c. Disaster from malicious programming or errors

d. Improved resource allocation

Correct Answer: c. Disaster from malicious programming or errors

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Which component is NOT mentioned as being responsible for allocating resources in the operating system?

a. File manager

b. Memory manager

c. Dispatcher

d. Scheduler

Correct Answer: c. Dispatcher

What information is stored in a process entry in the process table?

a. User's name and password

b. Priority of the process and memory area

c. Keyboard input data

d. Execution time of the process

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Correct Answer: b. Priority of the process and memory area

What is the primary purpose of an interrupt handler in the CPU?

a. Allocating resources to processes

b. Responding to external events and signals

c. Managing printer access

d. Controlling process priorities

Correct Answer: b. Responding to external events and signals

How does the CPU handle an interrupt signal?

a. It immediately resumes the interrupted task.

b. It saves its position in the current process and starts executing an interrupt handler.

c. It ignores the interrupt and continues with the current task.

d. It terminates the current process.

Correct Answer: b. It saves its position in the current process and starts executing an interrupt handler.

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What is the term for the procedure of changing from one process to another in a time-sharing system?

a. Process termination

b. Time slice completion

c. Process switch or context switch

d. Process initiation

Correct Answer: c. Process switch or context switch

How are multiprogramming CPUs designed to save a process's state when interrupted?

a. By immediately resuming the interrupted task

b. By storing the process's state in a separate memory area

c. By using machine-language instructions for state reloading

d. By creating a new process entry in the process table

Correct Answer: c. By using machine-language instructions for state reloading

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